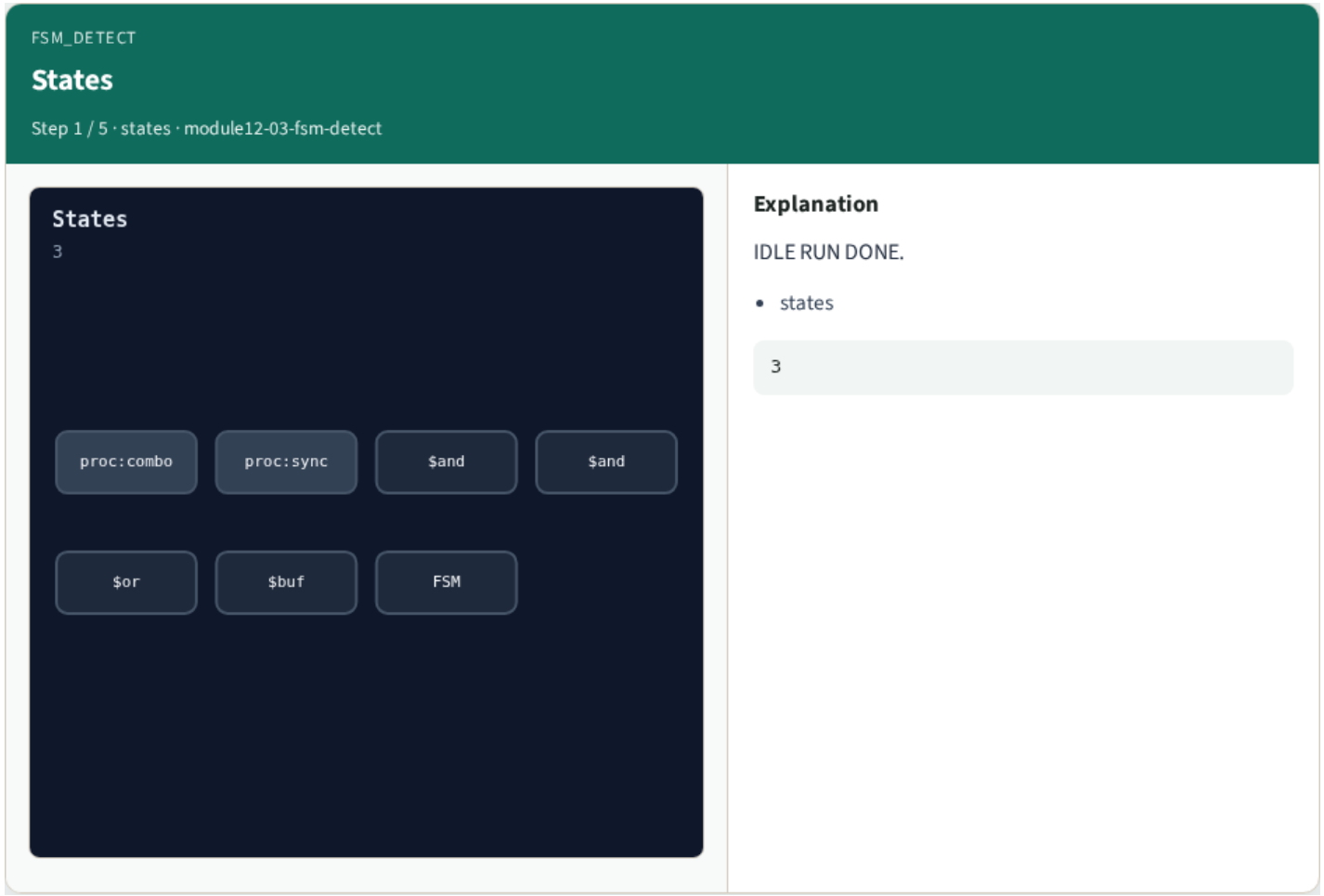
Why fsm

FSM passes count states and transitions, then choose encoding width

The idea

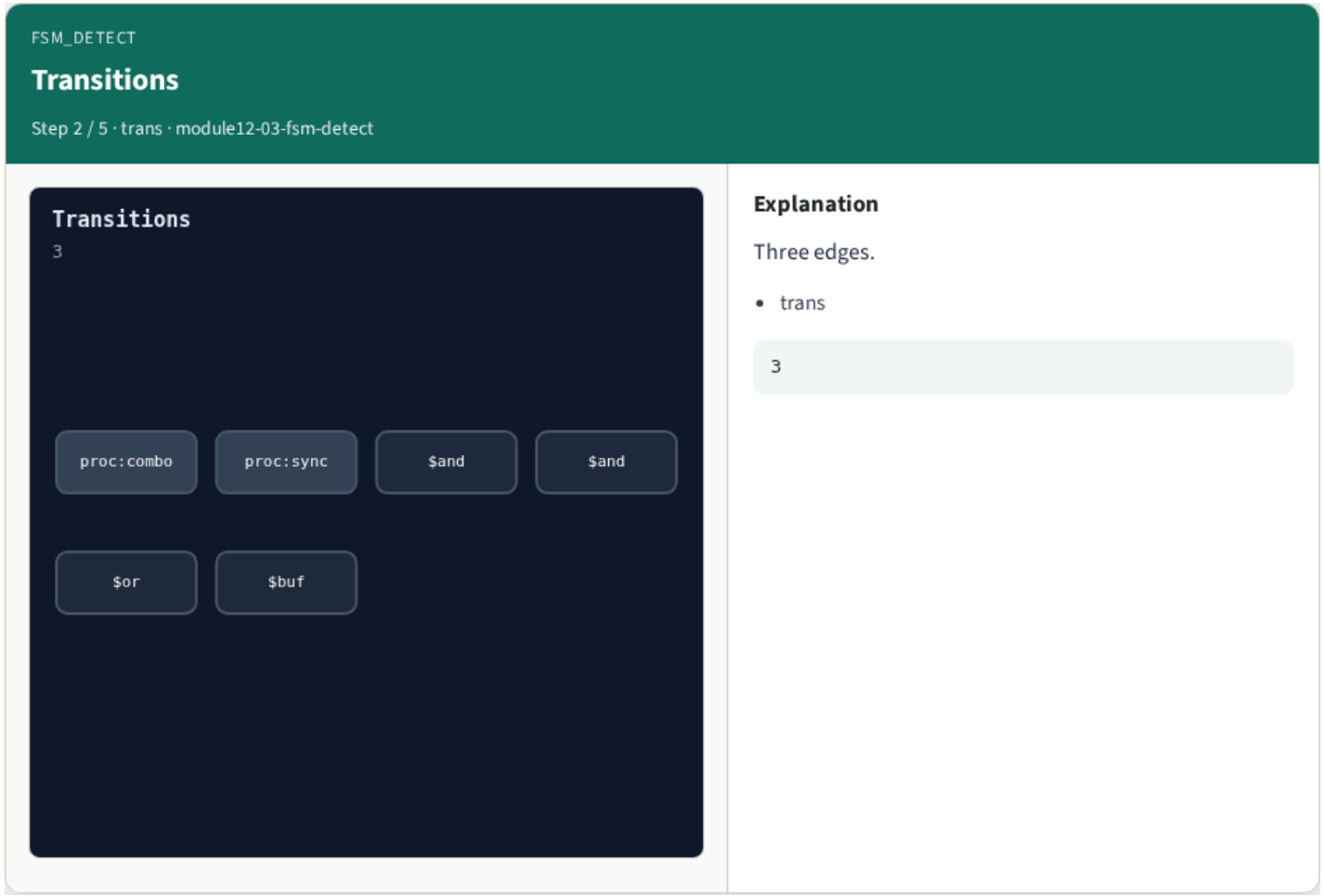
- GOLDEN: states=3; transitions=3; encoding_bits=2

States



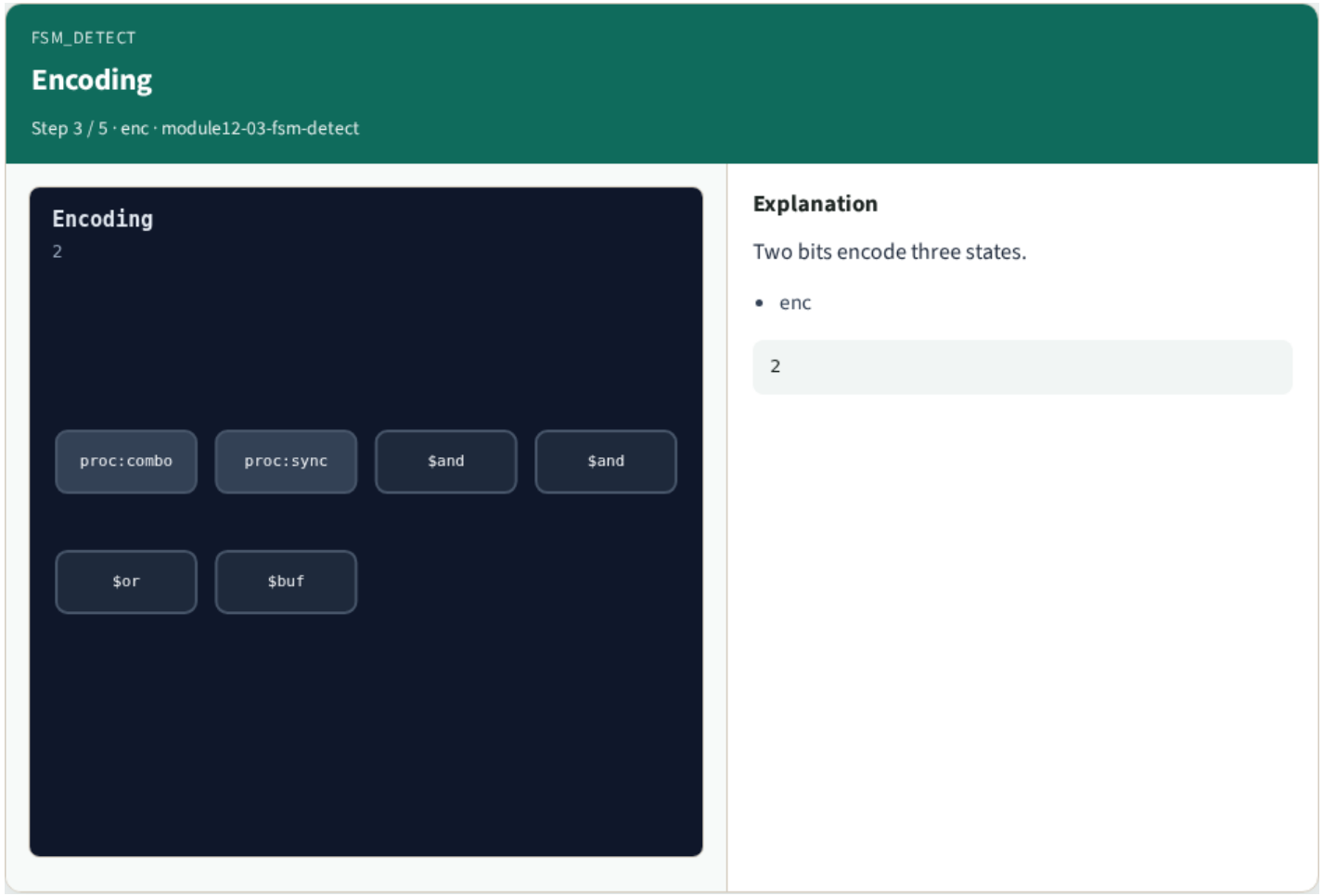
States

Transitions



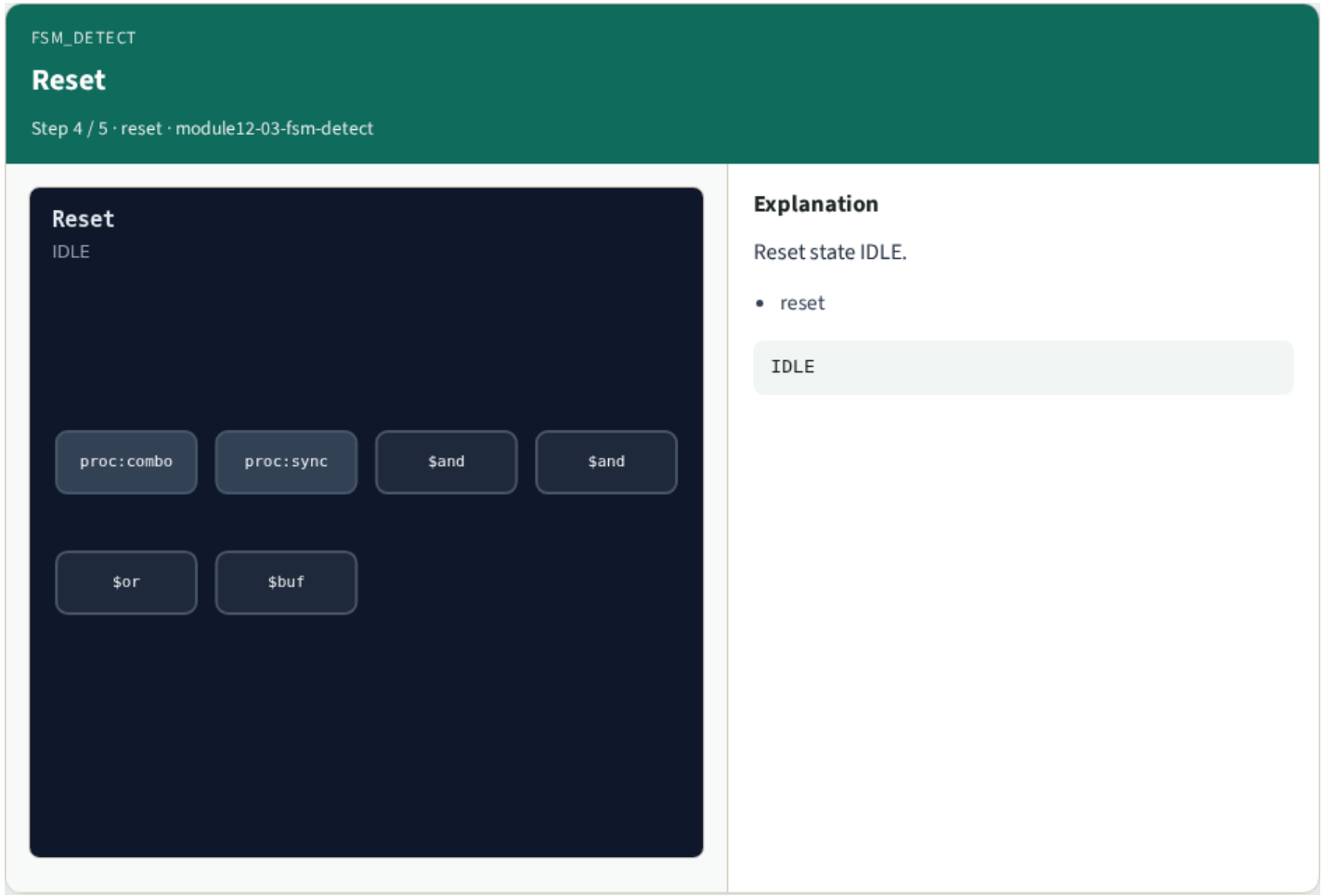
Transitions

Encoding



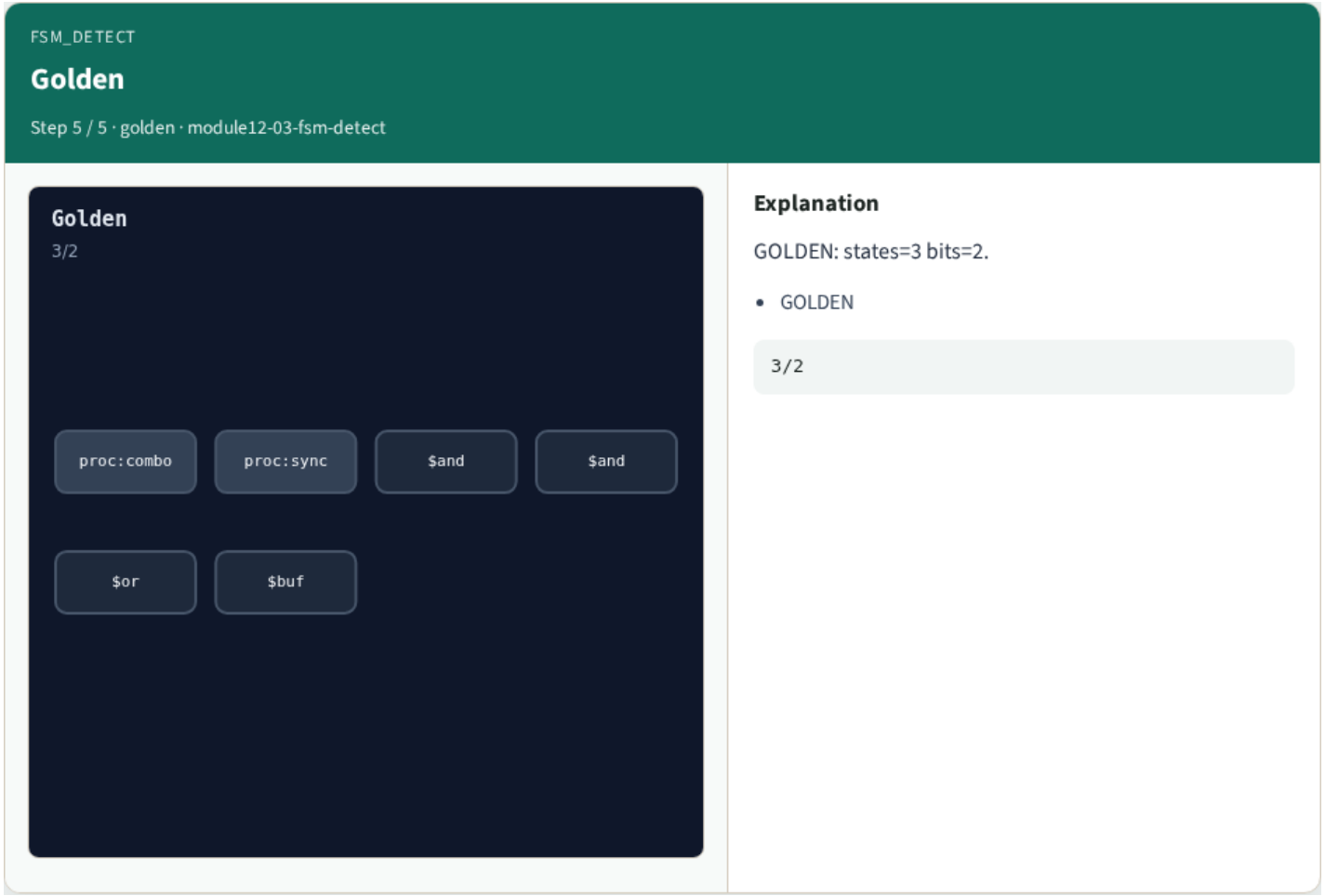
Encoding

Reset


FSM_DETECT
Reset
Step 4 / 5 · reset · module12-03-fsm-detect
Reset
IDLE
proc:combo
proc:sync
$and
$and
$or
$buf
Explanation
Reset state IDLE.
reset
IDLE

Reset

Golden



Golden

Browser lab track

- Open **fsm-detect**
- Read states, edges, and encoding width

Implement track

- Call algo_fsm_detect(); assert states == 3, transitions == 3, encoding_bits == 2

Pitfalls

- Counting encoding bits as state count
- Missing a transition edge

Try these

- From the common folder, run the algo solvers self-test, or call `algo_fsm_detect` directly

Try these — try these

```
cd courses/learn_yosys/common
python3 algo_solvers.py
```

Pseudocode

- Here is the check in plain steps

Pseudocode — try these

```
states ← fsm.states; trans ← fsm.transitions
bits ← max encoding string length
ok if 3, 3, and 2
```

Next

- Complete the quiz for this part
- Next: ABC handoff AND count